

**Faculty of Agribusiness and Commerce**

**Lincoln University**

**COMM 601 Principles of Managerial Economics**

**Assignment (Semester 1, 2026)**

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# **The Impact of Trade Protectionism and Exchange Rate Fluctuations: A Comparison of the New Zealand Textile Market and the Bangladesh Ready-Made Garment Industry**

## **Introduction**

International trade enables countries to specialise in the production of goods and services in which they have a comparative advantage. However, international trade can be influenced by exchange rate movements and government trade protection policies such as tariffs and quotas. **This report examines the textile and clothing market of New Zealand as the home country and the Ready-Made Garment (RMG) industry of Bangladesh as the foreign country.** New Zealand has relatively high labour and production costs, while Bangladesh benefits from lower labour costs and large-scale garment production (**BGMEA, 2025; Fashion & Textiles NZ, 2025**). The report analyses demand and supply conditions, price differences, exchange rate movements, trade gains and losses, and the effects of tariffs and quotas on the Home market.

## **Home Market: New Zealand Textile Industry**

The selected product for this report is a cotton T-shirt. New Zealand has a relatively small textile manufacturing sector compared with major garment-producing countries. The domestic textile industry focuses on specialised products such as wool, technical textiles, and premium clothing rather than mass-produced garments (**Fashion & Textiles NZ, 2025**).

Demand for cotton T-shirts in New Zealand is influenced by consumer income, preferences, population size, and fashion trends. When prices decrease, consumers are generally willing to purchase more T-shirts, while higher prices reduce the quantity demanded.

Supply in the New Zealand market is influenced by labour costs, production costs, technology, and business profitability. Because New Zealand has relatively high wages and operating costs, domestic manufacturers face higher production expenses than producers in Bangladesh.

The interaction between demand and supply determines the equilibrium price and quantity in the market (**OpenStax, 2023**).

The wider New Zealand fashion and textile industry contributes approximately NZD \$7.8 billion annually to the national economy and accounts for around 1.9% of New Zealand's GDP. The industry employs more than 76,000 workers and focuses primarily on premium wool products, technical textiles, and specialised manufacturing. Unlike major garment-producing countries, New Zealand relies heavily on imported clothing products because domestic labour and operating costs are relatively high. As a result, local producers often concentrate on niche markets rather than large-scale mass production (**Fashion & Textiles NZ, 2025**). Labour costs are one of the main factors contributing to higher production costs in New Zealand. Entry-level manufacturing workers earn approximately NZD \$23.95 per hour, while specialised textile workers and designers earn significantly higher salaries. These higher wages increase the overall cost of producing garments domestically and reduce the competitiveness of New Zealand manufacturers in low-cost clothing markets (**Figure NZ, 2025**).

#### **Foreign Market: Bangladesh Ready-Made Garment Industry**

Bangladesh is one of the world's largest exporters of ready-made garments. The industry contributes more than 80% of the country's export earnings and employs approximately four million workers directly. In 2025, Bangladesh's garment exports reached approximately US\$38.82 billion, making the country one of the leading suppliers of apparel products in the global market (BGMEA, 2025). The sector benefits from lower labour costs, large-scale manufacturing facilities, and extensive experience in export-oriented production.

The Bangladeshi garment industry produces a wide range of products including T-shirts, shirts, trousers, jackets, knitwear, and sportswear. Lower production costs enable Bangladeshi firms to supply international markets at competitive prices. Consequently, many retailers and clothing brands in developed countries source garments from Bangladesh (**World Bank, 2025**).

## Price Comparison Between Home and Foreign

New Zealand has significantly higher production costs than Bangladesh. Entry-level manufacturing wages in New Zealand are many times higher than garment-sector wages in Bangladesh. As a result, the retail price of a basic cotton T-shirt in New Zealand is generally higher than the price of a similar product manufactured in Bangladesh.

*For illustration:*

*New Zealand T-shirt price = NZD 10*

*Bangladesh T-shirt price = BDT 600*

Using the 2026 exchange rate (1 NZD = 71.93 BDT), the Bangladesh T-shirt price is approximately NZD 8.34. Although this price is higher than the artificial equilibrium price used in the demand and supply diagram (NZD 6), it still remains below the New Zealand market price and illustrates Bangladesh's cost advantage.

The lower cost structure in Bangladesh allows firms to produce garments at lower prices and export them competitively to international markets **(BGMEA, 2025)**.

**Table 1: Price Comparison**

| Country     | Product        | Average Price |
|-------------|----------------|---------------|
| New Zealand | Cotton T-shirt | NZD 10        |
| Bangladesh  | Cotton T-shirt | BDT 600       |

## Demand and Supply Diagrams for Home and Foreign Markets

To compare the markets of New Zealand and Bangladesh, demand and supply diagrams can be used. Since New Zealand has higher production costs, the equilibrium price is expected to be higher than in Bangladesh.

For the New Zealand market, assume the following demand and supply schedule:

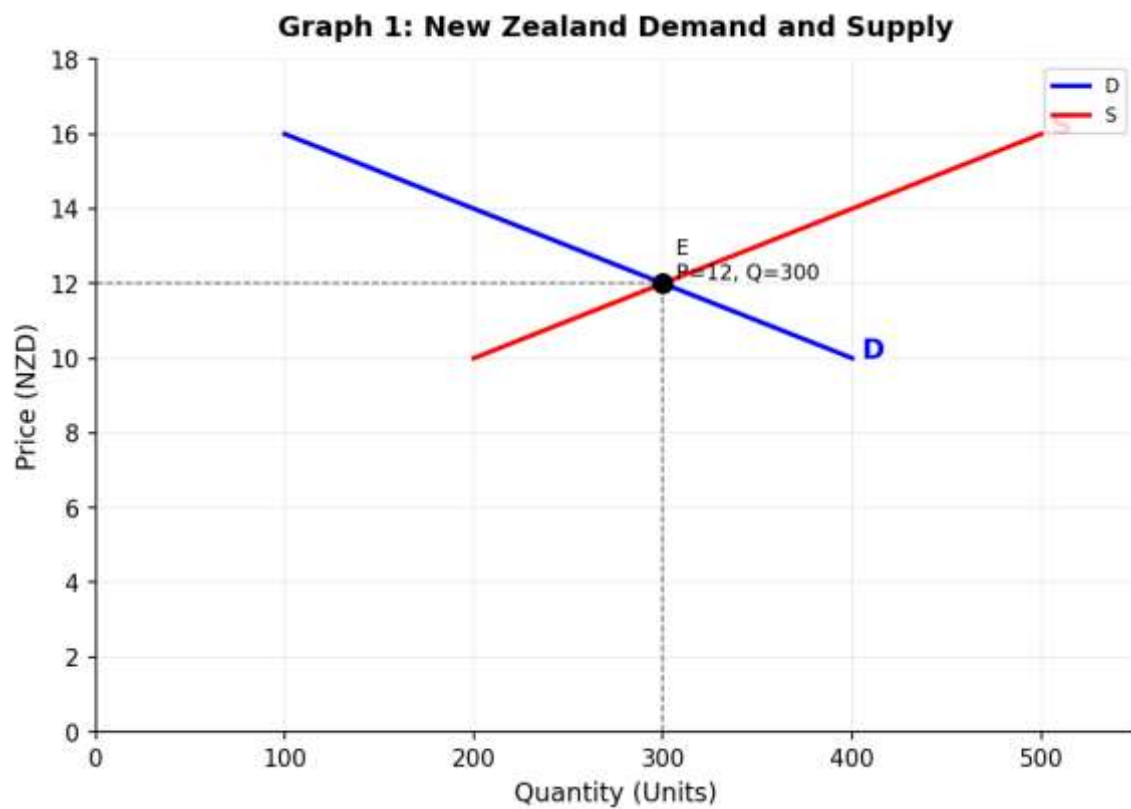
**Table 2: New Zealand Market**

| Price (NZD) | Quantity Demanded | Quantity Supplied |
|-------------|-------------------|-------------------|
| 16          | 100               | 500               |
| 14          | 200               | 400               |
| 12          | 300               | 300               |
| 10          | 400               | 200               |

The equilibrium occurs where quantity demanded equals quantity supplied.

*Equilibrium Price = NZD 12*

*Equilibrium Quantity = 300 units*



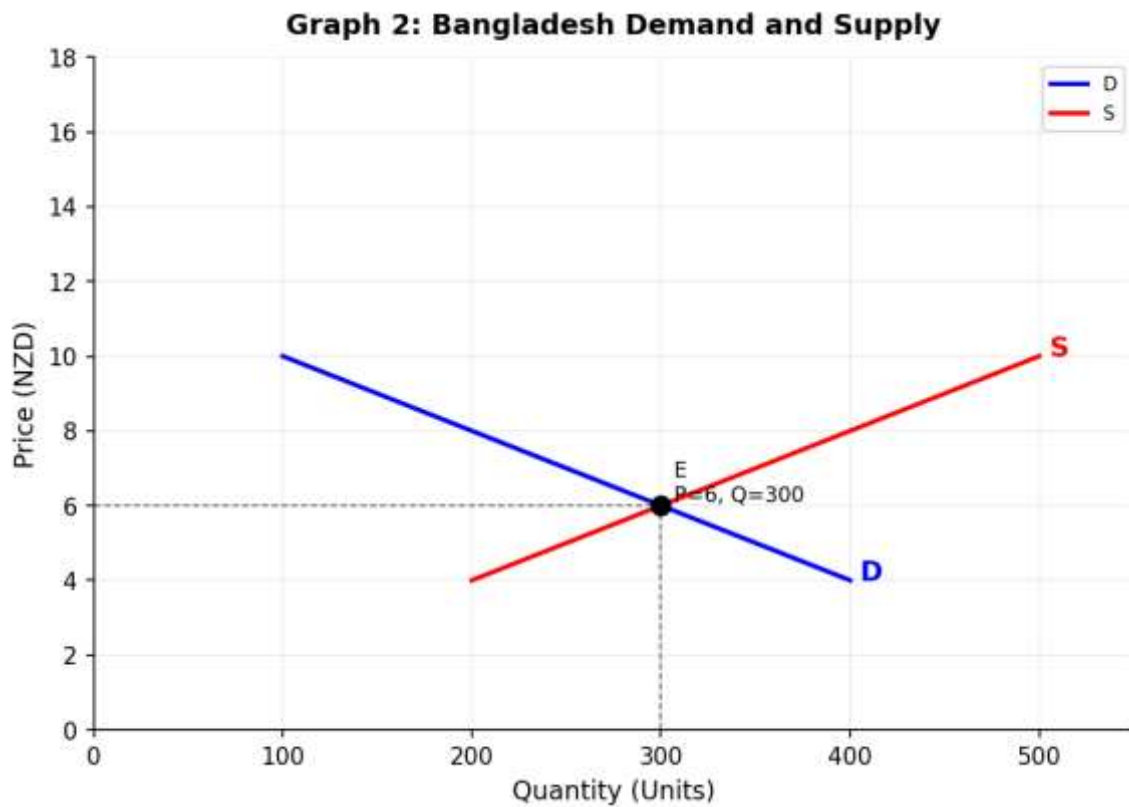
For the Bangladesh market, assume the following schedule:

**Table 3: Bangladesh Market**

| Price (NZD equivalent) | Quantity Demanded | Quantity Supplied |
|------------------------|-------------------|-------------------|
| 10                     | 100               | 500               |
| 8                      | 200               | 400               |
| 6                      | 300               | 300               |
| 4                      | 400               | 200               |

*Equilibrium Price = NZD 6*

*Equilibrium Quantity = 300 units*



The diagrams show that Bangladesh can supply the product at a lower equilibrium price than New Zealand due to lower labour and production costs (**OpenStax, 2023**).

## Nominal and Real Exchange Rates

Exchange rates play an important role in determining international competitiveness and trade flows between countries. The nominal exchange rate refers to the market exchange rate between two currencies and indicates how much of one currency can be exchanged for another.

**Table 4: Nominal Exchange Rate**

| Year | Exchange Rate (BDT per NZD) |
|------|-----------------------------|
| 2025 | 70.88                       |
| 2026 | 71.93                       |

The exchange rate increased slightly from 70.88 BDT per NZD in 2025 to 71.93 BDT per NZD in 2026 (**Wise, 2026**). This indicates a slight appreciation of the New Zealand dollar against the Bangladeshi taka. As a result, one New Zealand dollar could purchase more Bangladeshi taka in 2026 than in 2025.

However, the nominal exchange rate alone does not provide a complete picture of international competitiveness because inflation rates differ between countries. Therefore, the real exchange rate (RER) is calculated by adjusting the nominal exchange rate for differences in price levels.

### Real Exchange Rate Formula

Real Exchange Rate (RER) = Nominal Exchange Rate  $\times$  (Foreign Price Level  $\div$  Home Price Level)

For this analysis:

Bangladesh inflation rate: 10.47%

New Zealand inflation rate: 2.92%

(World Bank, 2025; Statistics New Zealand, 2025).

**Assuming a base-year price index of 100, the inflation rates are converted into price indices for real exchange rate analysis.**

Price Index (Bangladesh) = 110.47

Price Index (New Zealand) = 102.92

### **Real Exchange Rate Calculation**

$RER = 71.93 \times (110.47 \div 102.92)$

$RER = 71.93 \times 1.073$

$RER = 77.20$

The calculated real exchange rate of 77.20 indicates that after adjusting for inflation differences, Bangladeshi garments remain relatively competitive in the New Zealand market. Although Bangladesh experienced higher inflation than New Zealand during the study period, its significantly lower production costs and large-scale manufacturing capabilities helped maintain its comparative advantage in garment production.

Consequently, exchange rate movements and inflationary pressures did not eliminate Bangladesh's competitive position in the New Zealand market. Bangladeshi garments continued to be attractive to New Zealand importers because they could be supplied at lower prices than comparable products produced domestically in New Zealand (**Wise, 2026; World Bank, 2025**).

### **Comparative Advantage and International Trade**

International trade between New Zealand and Bangladesh is supported by the principle of comparative advantage. Bangladesh has a comparative advantage in garment production because it can manufacture clothing at a lower opportunity cost due to lower labour costs and a large workforce. In contrast, New Zealand specialises in products such as wool, dairy products, and technical textiles. As a result, Bangladesh can supply basic cotton T-shirts at lower prices than New Zealand producers (**OpenStax, 2023**).



Comparative advantage allows both countries to benefit from trade. New Zealand consumers gain access to lower-priced imported garments, while Bangladesh gains export revenue, employment opportunities, and foreign exchange earnings. This mutually beneficial relationship provides the economic justification for international trade between the two countries **(International Trade Centre, 2025)**.

### **Trade Gains and Losses in Home Market**

Without international trade, consumers in New Zealand purchase T-shirts at the domestic equilibrium price of NZD 12.

When international trade is allowed, New Zealand imports T-shirts from Bangladesh at NZD 6.

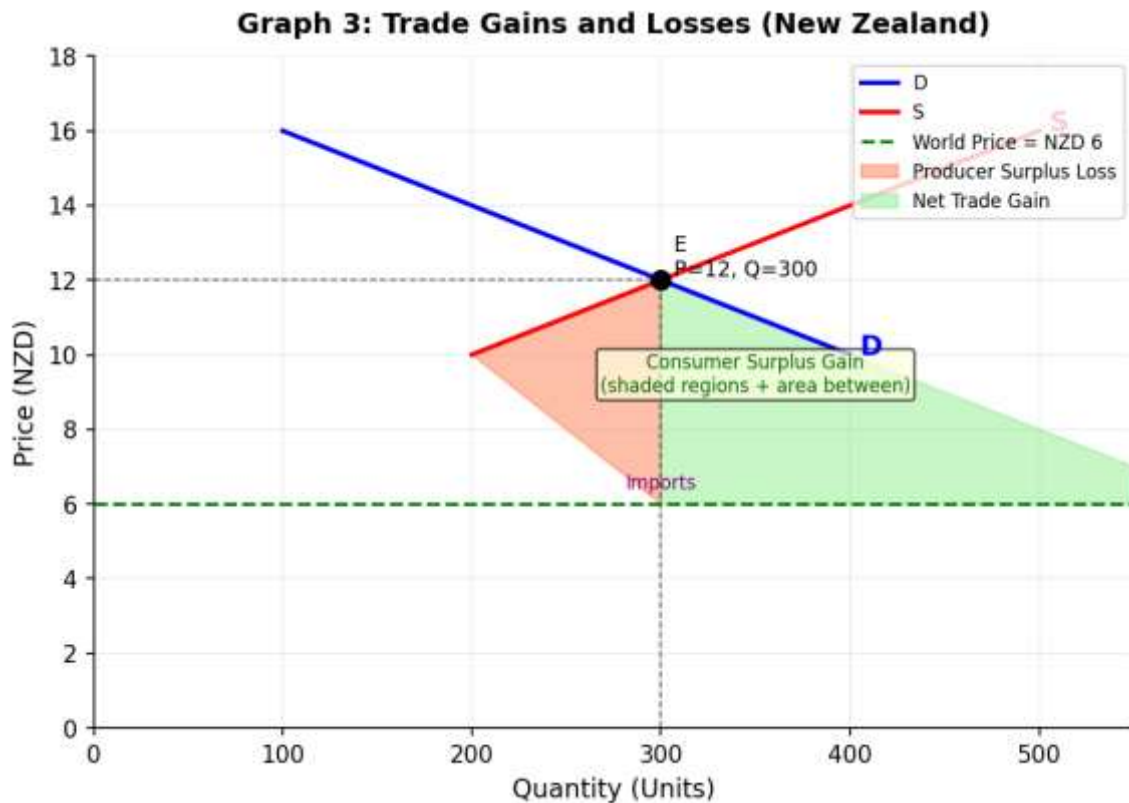
As a result:

- Consumers benefit from lower prices.
- Quantity demanded increases.
- Consumer surplus increases.

However:

- Domestic textile producers receive lower prices.
- Domestic production decreases.
- Producer surplus decreases.

Consumers therefore gain from trade, while some domestic producers lose market share **(OpenStax, 2023)**.



### Trade Protectionism: Tariffs and Import Quotas

Although international trade provides benefits to consumers through lower prices and greater product availability, governments may introduce protectionist measures to support domestic industries. The two most common trade protection instruments are import tariffs and import quotas.

#### 6.1 Tariff Scenario

A tariff is a tax imposed on imported products. Assume that imported Bangladeshi T-shirts enter New Zealand at a world price of NZD 6 per unit.

##### ***Scenario A: Low Tariff (Trade Continues)***

Assume the New Zealand government imposes a tariff of NZD 2 per imported T-shirt.

$$\text{Imported Price} = \text{World Price} + \text{Tariff}$$

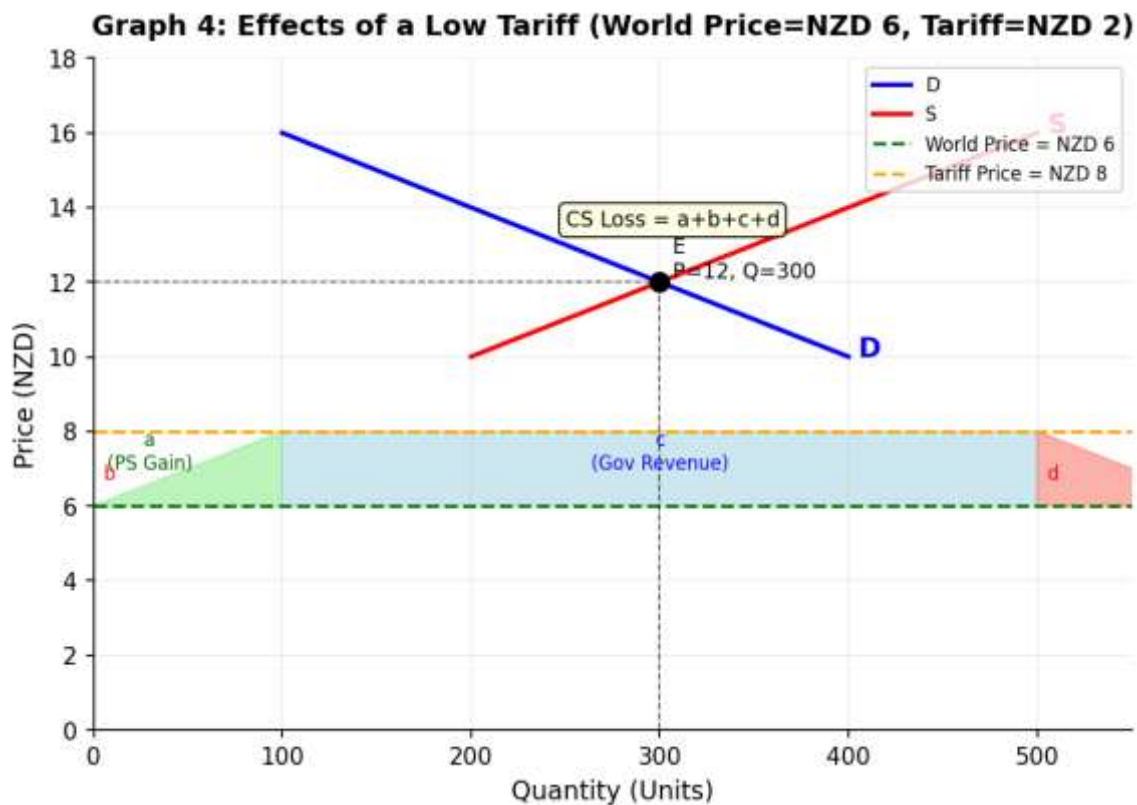
$$= 6 + 2$$

$$= \text{NZD } 8$$

As the imported price remains lower than the domestic equilibrium price of NZD 12, imports continue to enter New Zealand.

Effects:

- Consumers pay a higher price than under free trade.
- Consumer surplus decreases.
- Domestic producers receive a higher price and increase output.
- Government receives tariff revenue (**OpenStax, 2023**).



As shown in Graph 4, assume:

*Imports = 100 units*

*Tariff Revenue = Tariff  $\times$  Imports*

*=  $2 \times 100$*

*= NZD 200*

*Producer Surplus Gain is represented by area a = NZD 100*

*Government Revenue is represented by area c = NZD 200*

*Deadweight Loss is represented by areas b+d = NZD 50*

*Consumer Surplus Loss is represented by areas a + b + c + d = NZD 350*

### ***Scenario B: High Tariff (No Trade)***

Assume a tariff of NZD 6 per T-shirt.

*Imported Price*

$$= 6 + 6$$

$$= \text{NZD } 12$$

The imported price becomes equal to the domestic equilibrium price.

Consumers have little incentive to purchase imported products.

Imports effectively disappear (**OpenStax, 2023**).

Effects:

- Trade stops.
- Domestic producers regain market share.
- Consumer surplus decreases significantly.
- Government tariff revenue becomes negligible because imports effectively disappear.
- Deadweight loss increases because trade no longer occurs.

## **6.2 Import Quota Scenario**

An import quota limits the quantity of imported products.

### ***Scenario A: Quota Allows Trade***

***Assume New Zealand permits imports of 100 units.***

***Effects:***

- *Trade still occurs, but import volumes are restricted.*
- *Consumer surplus remains higher than under autarky but lower than under free trade.*
- *Domestic producers gain some protection and maintain market share.*
- *Market prices increase slightly compared with free trade.*
- *Import licences create quota rents for importers.*

**Assume:**

*Producer Surplus Gain = NZD 100*

*Quota Rent = NZD 100*

*Deadweight Loss = NZD 50*

*Consumer Surplus Loss*

*= Producer Surplus Gain + Quota Rent + Deadweight Loss*

*= 100 + 100 + 50*

*= NZD 250*

**Scenario B: Zero Import Quota**

Assume New Zealand imposes a quota of zero.

Effects:

- No imports are allowed.
- Domestic producers supply the entire market.
- Prices return to the domestic equilibrium level.
- Consumers lose access to lower-cost imported products.

The quota system protects domestic firms but reduces consumer welfare (**OpenStax, 2023**).

### 6.3 Comparison of Tariffs and Quotas

**Table 5: Comparison of Trade Protection Measures**

| Measure             | Trade Continues | Trade Stops |
|---------------------|-----------------|-------------|
| Low Tariff (NZD 2)  | Yes             | No          |
| High Tariff (NZD 6) | No              | Yes         |
| Quota (100 units)   | Yes             | No          |
| Quota (0 units)     | No              | Yes         |

Tariffs generate government revenue, whereas quotas primarily restrict imports without creating tax revenue.

## Currency Devaluation to Offset Tariffs

Suppose Bangladesh decides to offset the NZD 2 tariff imposed by New Zealand by devaluing its currency.

Before Devaluation

Exchange Rate:

$$1 \text{ NZD} = 71.93 \text{ BDT}$$

Bangladesh T-shirt Price:

$$600 \text{ BDT}$$

Price in NZD:

$$600 \div 71.93$$

$$= \text{NZD } 8.34$$

After Tariff:

$$8.34 + 2$$

$$= \text{NZD } 10.34$$

The tariff increases the final price faced by New Zealand consumers and reduces the competitiveness of Bangladeshi exports.

To fully offset the NZD 2 tariff, Bangladesh would need to devalue its currency so that the final post-tariff price remains equal to the original export price of NZD 8.34.

Required Exchange Rate:

Where E represents the exchange rate measured in Bangladeshi Taka (BDT) per New Zealand Dollar (NZD).

$$600 \div E + 2 = 8.34$$

$$600 \div E = 6.34$$

$$E = 600 \div 6.34$$

$$E = 94.64 \text{ BDT per NZD}$$

Therefore, Bangladesh would need to devalue its currency to approximately:

$$1 \text{ NZD} = 94.64 \text{ BDT}$$

Verification:

Price in NZD after devaluation:

$$600 \div 94.64$$

$$= \text{NZD } 6.34$$

After Tariff:

$$6.34 + 2$$

$$= \text{NZD } 8.34$$

The final price after the tariff remains equal to the original export price of NZD 8.34. Therefore, the effect of the tariff is fully offset by the currency devaluation. This demonstrates how exchange rate adjustments can help maintain export competitiveness despite the introduction of trade protection measures (Wise, 2026).

## Conclusion

This report examined the consequences of trade protectionism and exchange rate fluctuations using the New Zealand textile market as the Home country and the Bangladesh Ready-Made Garment industry as the Foreign country. The analysis showed that New Zealand faces higher production costs and higher domestic prices, while Bangladesh benefits

from lower labour costs and large-scale garment production. International trade allows New Zealand consumers to purchase imported garments at lower prices, generating gains in consumer welfare. However, domestic producers experience increased competition and lower producer surplus.

Tariffs and quotas can protect domestic industries by restricting imports, but they also increase prices for consumers and reduce overall economic efficiency. Furthermore, exchange rate movements significantly influence trade competitiveness. The analysis demonstrated that a depreciation of the Bangladeshi currency can partially offset the impact of tariffs and maintain export competitiveness in the New Zealand market. Overall, demand and supply conditions, comparative advantage, exchange rates, and trade protection policies all play important roles in shaping international trade outcomes.

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